National University of Sciences and Technology

School of Mechanical and Manufacturing Engineering

RAI-833 Deep Learning

Tutorial 5: Convolutional Neural Networks Report

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April 11, 2026

1 Introduction

In previous tutorials, image classification was performed using standard Artificial Neural Networks (ANNs). While ANNs can achieve high accuracy on simple datasets, they possess inherent limitations when processing images.

Why Transition to CNNs?

- **The ANN Limitation:** ANNs require flattening a 2D image into a 1D vector immediately, which destroys the spatial hierarchy and local relationships between adjacent pixels.
- **The CNN Solution:** Convolutional Neural Networks (CNNs) process images in their native 2D format. By utilizing learnable filters (kernels), CNNs perform **local feature extraction**—detecting edges, textures, and patterns directly from the spatial data.

This report details the implementation and evaluation of a CNN architecture designed to classify the MNIST handwritten digit dataset efficiently.

2 Methodology

The CNN was implemented using the PyTorch framework. The architecture consists of a feature extraction block followed by a classification block.

2.1 Network Architecture

- **Conv2D Layer:** The input grayscale image ($28 \times 28 \times 1$) is passed through a convolutional layer with **32 filters** and a 3×3 **kernel**, utilizing a ReLU activation function. This extracts fundamental visual features.
- **MaxPooling2D Layer:** A 2×2 **pooling operation** is applied to downsample the spatial dimensions from 26×26 to 13×13. This drastically reduces computational load and provides spatial translation invariance.
- **Flatten and Dense Layers:** The resulting 3D tensor is flattened into a 1D vector of **5,408 features** ($32 \times 13 \times 13$). This vector is passed into a fully connected (Linear) layer of **100 neurons** with ReLU activation, and finally to an output layer of **10 neurons** corresponding to the target digit classes.

2.2 Training Setup

The network was trained using the following configuration:

- **Optimizer:** `Adam`
- **Loss Function:** `CrossEntropyLoss`
- **Epochs:** 5
- **Batch Size:** 64

3 Results and Discussion

The CNN converged rapidly, demonstrating exceptional learning efficiency. By the end of the 5th epoch, the **training accuracy reached 99.16%** alongside a highly stable **validation accuracy of 98.48%**.

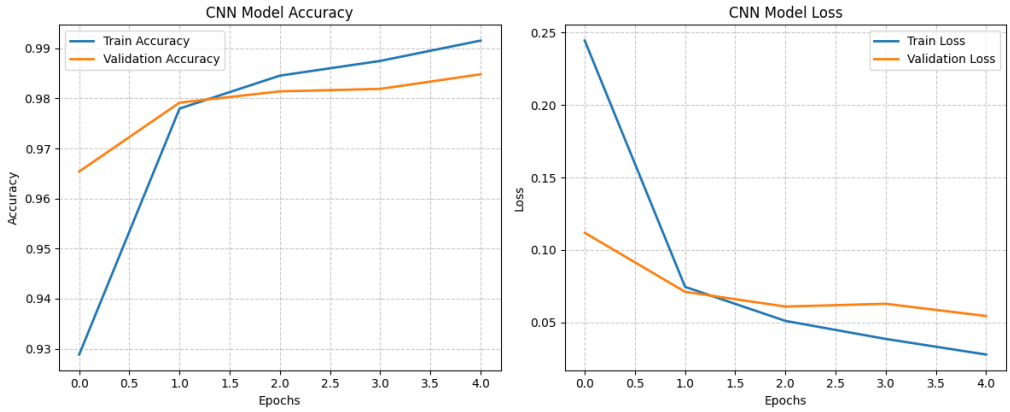

Figure 1: Training and Validation Accuracy/Loss curves across 5 epochs.

3.1 Test Set Evaluation

When evaluated on the unseen MNIST test set, the final model achieved an outstanding overall accuracy of **98.28%**. The detailed performance metrics across all ten digit classes are summarized in Table 1.

Table 1: Classification Report on MNIST Test Set

Digit Class	Precision	Recall	F1-Score	Support
0	0.9750	0.9959	0.9854	980
1	0.9947	0.9921	0.9934	1135
2	0.9816	0.9806	0.9811	1032
3	0.9861	0.9861	0.9861	1010
4	0.9907	0.9766	0.9836	982
5	0.9811	0.9877	0.9844	892
6	0.9937	0.9802	0.9869	958
7	0.9686	0.9893	0.9788	1028
8	0.9704	0.9774	0.9739	974
9	0.9858	0.9613	0.9734	1009
Accuracy			**0.9828**	**10000**
Macro Avg	0.9828	0.9827	0.9827	10000
Weighted Avg	0.9829	0.9828	0.9828	10000

3.2 Visual Predictive Analysis

To further analyze the model's predictive robustness, visual diagnostics were generated:

- **Confusion Matrix (Figure 2):** Indicates minimal misclassifications across all target classes, proving the model rarely confuses one specific digit for another.

- **Sample Predictions (Figure 3):** Visually confirms the network's ability to accurately classify diverse handwriting styles from the test set.

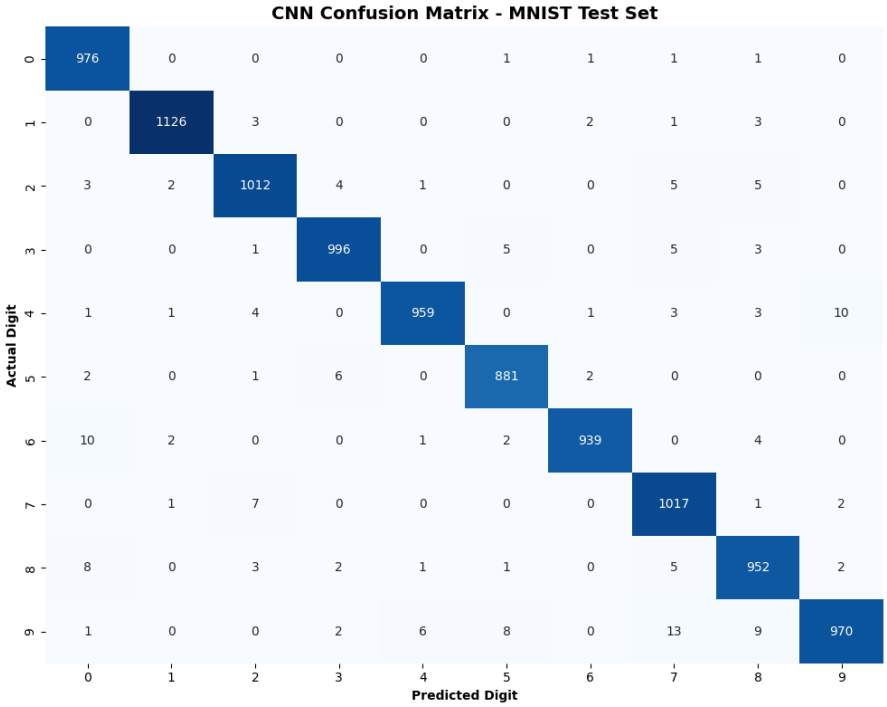

Figure 2: Confusion Matrix evaluated on the final unseen test set.

Figure 3: A 3x3 grid of random test images displaying true and predicted labels.

4 Conclusion

The CNN architecture proved to be highly effective for the MNIST classification task.

Key Takeaways:

- **High Efficiency:** The model achieved a remarkable final test accuracy of **98.28%** in just 5 epochs.
- **Hierarchical Learning:** By leveraging `Conv2D` layers for local spatial feature extraction and `MaxPooling` to reduce dimensionality, the model learned visual representations that a standard Multi-Layer Perceptron simply cannot capture.
- **Optimal Architecture:** The results confirm that Convolutional Neural Networks represent the optimal, state-of-the-art approach for analyzing grid-like topology data such as images.